

Introduction to Lie Group Theory

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Chapter 1

Real and Complex Matrix Groups

1.1 Groups of Matrices

1.2 Complex Matrices as Real Matrices

1.3 Continuous Homomorphisms of Matrix Groups

1.4 Matrix Groups of Normed Vector Spaces

1.5 Continuous Group Actions

Chapter 2

Exponentials, Differential Equations, and One Parameter Subgroups

2.1 Matrix Exponential and Logarithm

2.2 Differential Equations in Matrices

2.3 One Parameter Subgroups in Matrix Groups

Chapter 3

Tangent Spaces and Lie Algebras

3.1 Lie Algebras

3.2 Curves and Tangent Spaces

3.3 Lie Algebras of Matrix Groups

3.4 $SO(3)$ and $SU(2)$

3.5 Complexification of Real Lie Algebra

Chapter 4

Lie Groups

4.1 Smooth Manifolds

4.2 Tangent Spaces and Derivatives

4.3 Lie Groups

4.4 Matrix Groups are Lie Groups

4.5 Not All Lie Groups are Matrix Groups

References

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